

Lab 3: Introduction to Vectors

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, September 16th, 2026

Name: _____

username: _____

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all activities and show your lab TA by the end of the lab section.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

In each category, select the option that best describes your (honest) thoughts towards this class so far.

i. Difficulty

- ☐ Way harder than I expected
- ☐ Harder than I expected
- ☐ About as hard as I expected
- ☐ Easier than I expected
- ☐ Way easier than I expected

ii. Interest

- ☐ Way more interesting than I expected
- ☐ More interesting than I expected
- ☐ About as interesting as I expected
- ☐ Less interesting than I expected
- ☐ Way less interesting than I expected

Any immediate action items we can take to make your experience better?

Activity 1: Transformed Data

Let's begin by reviewing how transforming a dataset affects a fitted simple linear model.

Suppose we're given a dataset of n points, $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, where $\bar{x}$ is the mean of $x_1, x_2, \dots, x_n$ and $\bar{y}$ is the mean of $y_1, y_2, \dots, y_n$.

Using this dataset, we create a *transformed* dataset of n points, $(x'_1, y'_1), (x'_2, y'_2), \dots, (x'_n, y'_n)$, where:

$$x'_i = 4x_i - 3 \quad y'_i = y_i + 24$$

So the transformed dataset is of the form

$$(4x_1 - 3, y_1 + 24), (4x_2 - 3, y_2 + 24), \dots, (4x_n - 3, y_n + 24)$$

We decide to fit a simple linear model $h(x'_i) = w_0 + w_1 x'_i$ on the transformed dataset using squared loss. We find that $w_0^* = 7$ and $w_1^* = 2$, so $h^*(x'_i) = 7 + 2x'_i$.

- a) Suppose we were to fit a simple linear model through the original dataset, again using squared loss. What would the optimal slope on the original dataset be?

- b) Recall, the model $h^*(x'_i) = w_0 + w_1 x'_i$ was fit on the transformed dataset, introduced above. $h^*(x'_i)$ happens to pass through the point $(\bar{x}, \bar{y})$. What is the value of $\bar{x}$? Give your answer as an integer with no variables. *Hint: What else does $h^*(x'_i)$ pass through?*

Recap: Vectors

Now that we've reviewed simple linear regression, let's turn to vectors. The next three activities explore vector lengths, linear combinations, and their implementation in NumPy.

- ([Chapters 3.1](#) and [3.2](#)) The **norm** of a vector $\vec{v} \in \mathbb{R}^n$ measures its length:

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

This is the default norm for vectors in $\mathbb{R}^n$, but other norms exist.

- ([3.1](#)) A **linear combination** of the vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$ is any vector that can be written as

$$a_1\vec{v}_1 + a_2\vec{v}_2 + \cdots + a_d\vec{v}_d$$

where $a_1, a_2, \dots, a_d$ are scalars. We can think of this as taking bits of each vector and adding them together. The a_i 's are called the **coefficients** of the linear combination.

Activity 2: Triangle Inequality

We first use norms to compare the length of a sum with the lengths of the original vectors. The triangle inequality states that for any two vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$:

$$\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$$

- a) For the vectors $\vec{u} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} -1 \\ -3 \end{bmatrix}$, verify that the triangle inequality holds. That is, show that the left-hand side is less than or equal to the right-hand side.

- b) Find two **different** vectors $\vec{x}, \vec{y} \in \mathbb{R}^2$ such that the triangle inequality achieves **equality**, i.e. where

$$\|\vec{x} + \vec{y}\| = \|\vec{x}\| + \|\vec{y}\|$$

What is the relationship between the $\vec{x}$ and $\vec{y}$ you found?

Activity 3: Linear Combinations

Next, we'll look at which vectors we can form by scaling and adding other vectors.

Let $\vec{u} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} -1 \\ -3 \end{bmatrix}$, and $\vec{w} = \begin{bmatrix} -6 \\ 9 \end{bmatrix}$.

- a) Find values of a and b such that $a\vec{u} + b\vec{v} = \vec{w}$. By finding a and b , you have written $\vec{w}$ as a **linear combination** of $\vec{u}$ and $\vec{v}$.

- b) Now, try and write $\vec{w}$ as a linear combination of $\vec{u}$, $\vec{v}$, and $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$. In other words, try and find values of a , b , and c such that

$$a \begin{bmatrix} 4 \\ 3 \end{bmatrix} + b \begin{bmatrix} -1 \\ -3 \end{bmatrix} + c \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \vec{w}$$

What happens? Why?

- c) Now, try and write $\vec{w}$ as a linear combination of $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -4 \\ -2 \end{bmatrix}$. What happens? Why?

Activity 4: Arrays in NumPy

Now, put vector operations into practice using NumPy arrays. Instead of writing code in a separate Jupyter Notebook for this lab, you will interact with the code cells that exist in the course notes.

In particular, go to [Chapter 3.2](#) of the course notes, scroll all the way to the bottom, and complete **Activity 5** there. To get checked off, show your lab TA that you've completed the activity — there's no need to submit your code anywhere.

Introduction: The Dot Product

So far, we've measured vector lengths and formed linear combinations. The dot product gives us a way to compare the directions of two vectors. We will study this in tomorrow's lecture; use the definitions below to explore it in the next two activities.

- (3.3) The **dot product** of two vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$ is defined as:

$$\vec{u} \cdot \vec{v} = \begin{bmatrix} u_1 \\ u_2 \\ \dots \\ u_n \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

The result is a **scalar**, not another vector.

- (3.3) The dot product also has a geometric definition, involving the norms (lengths) of two nonzero vectors and the angle between them:

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$$

- (3.3) The key takeaway from the dot product is that it tells us how similar the directions of two vectors are. When two vectors have a dot product of 0, they are **orthogonal**. For two nonzero vectors, this means that the angle between them is 90° .

Activity 5: The Dot Product

For each pair of vectors below, (1) draw them on the grid to the right and (2) compute their dot product using the component-wise definition.

a) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$



b) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ 0 \end{bmatrix}$



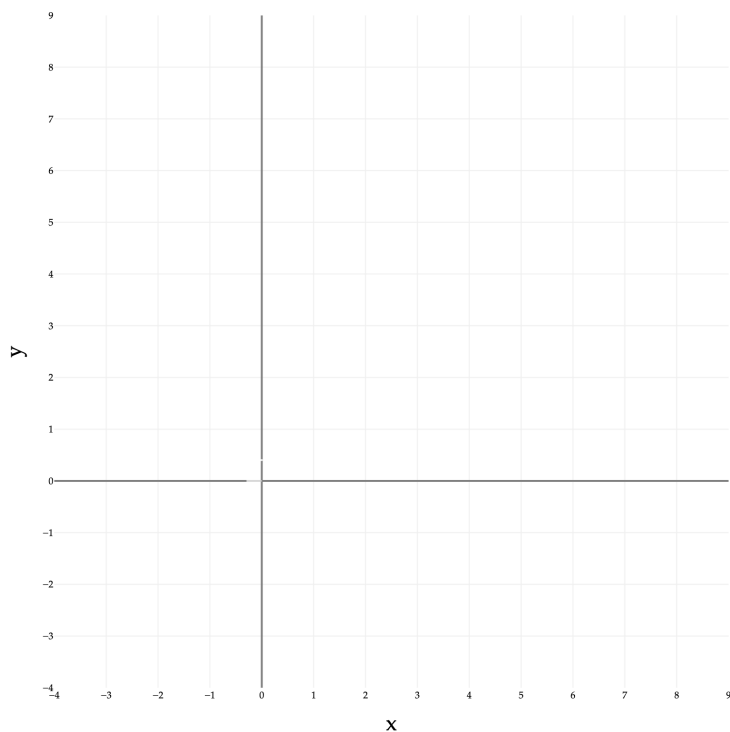
c) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 6 \\ 8 \end{bmatrix}$



d) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$



e) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ 4 \end{bmatrix}$



Activity 6: Angles and Orthogonality

In the previous activity, we computed dot products using vector components. Now, we will connect those calculations to angles using the geometric definition of the dot product.

$$\vec{u} \cdot \vec{v} = \begin{bmatrix} u_1 \\ u_2 \\ \dots \\ u_n \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix} = u_1v_1 + u_2v_2 + \dots + u_nv_n$$

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$$

Let $\vec{w} = \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \end{bmatrix}$ and $\vec{x} = \begin{bmatrix} 9 \\ 1 \\ 2 \\ 3 \end{bmatrix}$.

- a) Find $\vec{w} \cdot \vec{x}$, $\|\vec{w}\|$, and $\|\vec{x}\|$.

- b) Using the results of part a), find the angle between $\vec{w}$ and $\vec{x}$. Leave your answer in the form $\cos^{-1}(\cdot)$.

- c) What is $\cos(90^\circ)$? What does this have to do with orthogonality?